



OptiTemp® – Batteries

A458935

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Operator's Manual

A465990 rev 1

Contents

1. FOREWARD	3
1.2 Warning symbols used in this manual	3
1.3 Important relation to other safety documents	4
2. DESIGNED PURPOSE OF USE.....	4
3. SAFETY INSTRUCTIONS	5
3.1 Personal protective equipment	6
4. TECHNICAL SPECIFICATION.....	6
4.1 Safety labels, main components and dimensions of the battery.....	7
5. REMOVING OR INSTALLING THE BATTERY	10
6. CONNECTING AND DISCONNECTING THE CHARGER CONNECTOR	10
7. OPERATION OF THE BATTERY BY MULTI-FUNCTION DISPLAY.....	10
8. SAFETY FEATURES OF THE BATTERY MANAGEMENT SYSTEM	11
9. ROUTINE INSPECTIONS BEFORE USE OF THE E-LOADER.....	11
10. MAINTENANCE AND SERVICE	12
11. WARRANTY TERMS	12
12. EC DECLARATION OF CONFORMITY	13
12.1 Norms & regulations	13
13. TRANSPORT INFORMATION	13

1. FOREWORD

By familiarizing yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the battery. Read the instructions carefully before starting to use the battery or performing maintenance.

The purpose of this manual is to help you to:

- operate the battery in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the battery intact and ensure a long service life

Using the battery for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from your Avant dealer or from the manufacturer.

In addition to the safety instructions included in this Operator's Manual, you must observe all occupational safety regulations, local laws, and other regulations concerning the use of the equipment.

This manual contains the original instructions in English. Due to continuous product development some of the details shown in this manual may differ from your equipment. Pictures may also depict optional equipment or features that are not currently available. We reserve the right to change the contents of the manual without notification.

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1.2 Warning symbols used in this manual

The following warning symbols are used throughout this manual. They indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:



WARNING SAFETY ALERT SYMBOL

This symbol means: **"Warning, be alert! Your safety is involved!"**

Carefully read the message that follows, it warns of an immediate hazard that could cause serious personal injury.

The safety alert symbol by itself and with related safety statement indicates important safety messages throughout this Manual. It is used to draw attention to instructions involving your personal safety or the safety of others. When you see this symbol, be alert, your safety is involved, carefully read the message that follows, and inform other operators.

DANGER

This signal word indicates a hazardous situation which, if not avoided, will cause death or serious injury.

WARNING

This signal word indicates a potentially hazardous situation which, if not avoided, could cause serious injury or death.

CAUTION

This signal word is used when minor injury could result if the instructions are not followed properly.

NOTICE

This signal word indicates information about the correct operation and maintenance of the battery.

Failure to observe the instructions accompanying the symbol can lead to battery breakdown or other property damage.

1.3 Important relation to other safety documents



Read carefully A466140 Battery safety procedure, which describes in detail hazards and safety precautions.

Read carefully A466139 Optitemp Fluid Safety Datasheet, which describes in detail substance chemical hazards and safety precautions.

2. DESIGNED PURPOSE OF USE

The AVANT POWER Optitemp LiION battery pack is rechargeable energy source for use with ELECTRIC AVANT LOADERS. The battery pack is intended to provide energy to drive and operate the Loader.

Optitemp® LiION battery packs can be delivered with two different sizes. Technical specifications and battery pack models are shown in Table 1. in chapter 4.

3. SAFETY INSTRUCTIONS

**DANGER**

Misuse, careless use, or using a battery which is in poor condition may cause risk of serious injuries. Familiarize yourself with the proper functions of the loader's display, correct visual inspection and cleanliness of the battery. Read all safety precautions carefully.

Read all safety instructions carefully before handling the e-Loader

**WARNING**

- **If the battery is exposed to a fire, hazardous substances from the battery can be released, resulting in a risk of battery fire and/or harmful gas emissions.** Avoid driving near firework areas, fireplaces or other open fire. The battery housing is designed to resist temperature up to 80°C.
- **Risk of fire and electric shock in the battery if the mechanical protection of the battery system is penetrated causing internal short circuit and/or thermal propagation.** Always make sure to visually check the working area or driving route for other vehicles, such as forklifts or objects including spikes. The loader's steel body does not totally protect the battery housing for penetrating objects.
- **Crush or collision of the Loader causes a risk of battery housing rupture, resulting fire and/or harmful gas emissions.** Always focus on driving the Loader carefully. In addition, always observe the surrounding areas for other vehicles or objects. The battery is protected by Loader steel body, to withstand collision or crush with designed driving speed.
- **Risk of electric shock and thermal liquid leak if the battery drops when removing from or installing it into a Loader.** Lifting device is required. Do not remove or install the battery yourself. Only Avant service is authorized to remove or install a battery.
- **Risk of fire and electric shock if the charger is damaged, incorrect connection of the charger or incorrect charger is used.** Always check that the connection cables and connectors are clean and not damaged. Do not disconnect during charging. When charging is finished or stopped by the user, it is safe to disconnect the charger.
- **Risk of electric shock, fire, and thermal liquid leakage if the Loader is subjected to submersion and battery housing is damaged.** Avoid driving areas where it is possible to submerge the loader or battery in water. Regularly visually inspect that the battery housing is not damaged. Battery housing is designed to prevent water leaking in when the housing is not damaged.

NOTICE

REGULARLY INSPECT that the Loader's battery mounting space is free from water, dirt, dust or other external objects.

3.1 Personal protective equipment

Remember to wear proper personal protective equipment:



- Wear protective gloves when cleaning the battery and mounting space.



- Wear safety glasses e.g. when using cleaning aids.

4. TECHNICAL SPECIFICATION

Table 1 - Specifications

Battery Model	Double-module OptiTemp® battery	4-module OptiTemp® battery
Voltage	44V	44V
Capacity	13 kWh	27 kWh
Module Ah _{nom}	260 Ah	520 Ah
Weight	120 kg	200 kg
IP-class	54	
Housing material	PE+PP	
Heating and cooling	Optitemp Fluid	
Quantity of Modules	2	4
Discharging temperature	-20 ... +60 °C	
Charging temperature	0 ... +45 °C	
Storage temperature	0 ... +25 °C	
Storage time	6 months	
Dimensions	See section 4.1	
Main components	See section 4.1	
Charging connectors, AC	Onboard AC charger included (TYPE2)	
Rapid charging connector, DC	Rapid charging connector included, charger is optional	
Cell Model	NCR2170M	
Cells quantity	750	1440
Cell Rated capacity	4920 mAh	
Cell typical energy	18.8 Wh	
Cell nominal voltage	3.6 V	
Cell maximum allowed charge current	2510 mA	
Cell maximum allowed discharge current	12 A	

4.1 Safety labels, main components and dimensions of the battery

Listed below are the labels and markings attached to the battery and e-Loader. They must be visible and readable on the equipment. Replace any unclear or missing label. New labels are available via your retailer or contact information provided on the cover.



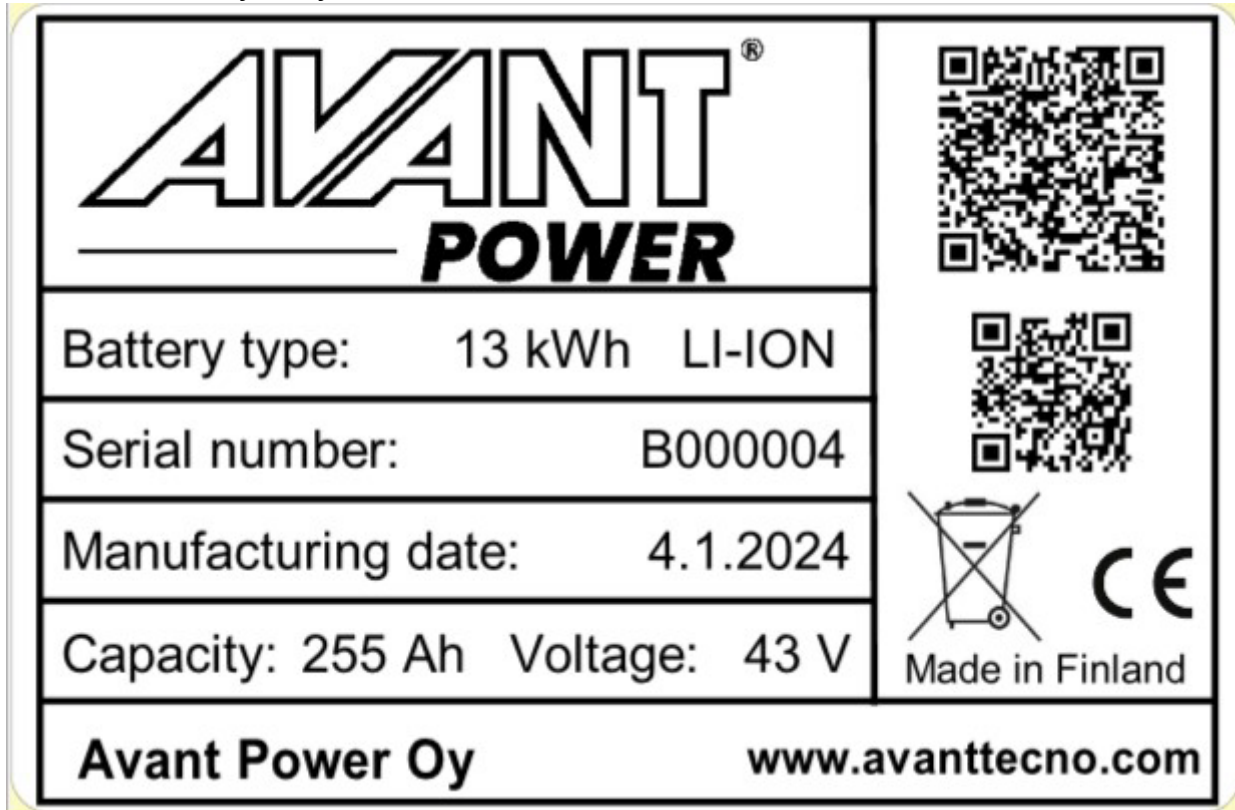
WARNING

Make sure all warning decals are readable. The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.

Picture 1. – Safety labels attached on the battery or near of it



Picture 2 – Example of identification label

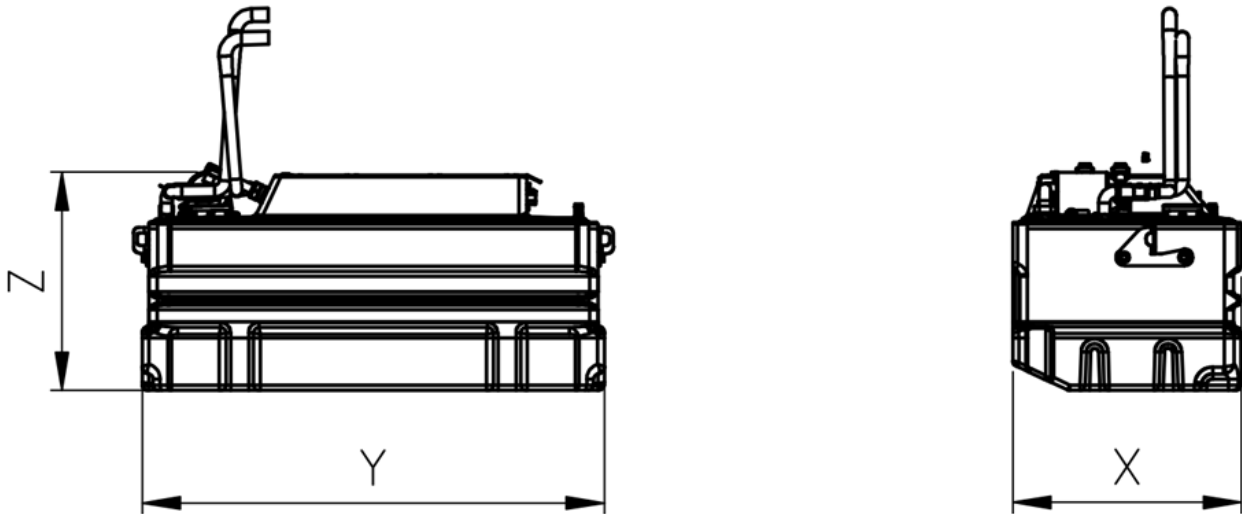


Picture 3 - Warranty label

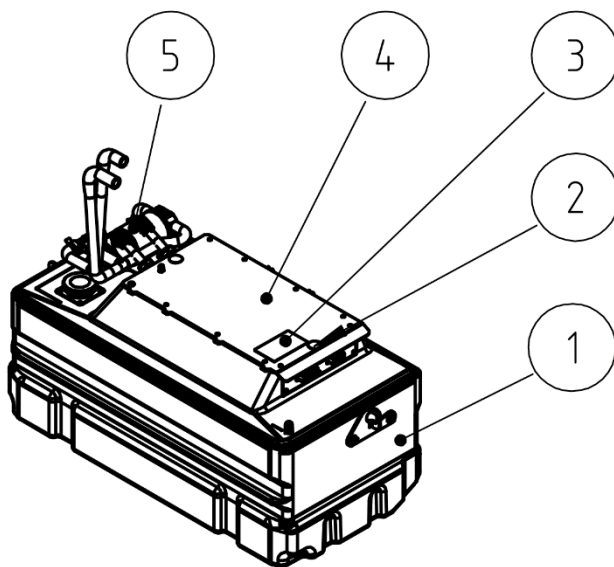


Picture 4 – Dimensions of the batteries

Battery Model	Width (X)	Length (Y)	Height (Z)
Double-module OptiTemp battery	419 mm	846 mm	400 mm
4-module OptiTemp battery	419 mm	846 mm	567 mm



Picture 5 – Main components of the batteries



1. Battery housing
2. Warranty label
3. Safety label
4. BMS section
5. Thermal pump

5. REMOVING OR INSTALLING THE BATTERY

NOTICE

Do not remove or install the battery yourself. Only Avant service is authorized to perform such maneuvers. Contact your local Avant service provider for further actions.

6. CONNECTING AND DISCONNECTING THE CHARGER CONNECTOR



- **Risk of fire and electric shock if the charger is damaged, incorrect connection of the charger or incorrect charger is used.** Always visually observe the connection cables and connectors are clean and not damaged. Do not disconnect during charging. When charging is finished or stopped by the user, it is safe to disconnect the charger.

Battery can be charged either by onboard charger (AC charging) or by rapid charger (DC charging). Rapid charger is optional.

There are two connector types in the loader; onboard AC charger TYPE2 connector and DIN 320 DC connector for optional rapid charging.

When charging the battery, TYPE2 charging station is required. To start charging, connect TYPE2 cable to TYPE2 socket, after some seconds battery will lock TYPE2 cable and start charging. If locking fails, battery informs this by blinking unlocking button LED, then remove TYPE2 cable from socket and try connecting it again. To stop or remove locked TYPE2 cable from socket, press unlocking button, battery will stop charging and opens TYPE2 cable locking, you can remove the cable when all LEDs are turned off. If unlocking fails, battery informs this by blinking unlocking button LED. To retry unlocking press unlocking button again after 5 seconds.

If the Loader is charged with rapid charger station (DC), charging starts when connecting the cable to socket in Loader. To stop charging, first press stop button in charger station and wait until the station informs charging ended and then remove charger connector. Never remove charging cable when charging is active.

For more details, please read the Operator's manual of the loader and Operator's manual of the rapid charger.

7. OPERATION OF THE BATTERY BY MULTI-FUNCTION DISPLAY

The multi-function display of the loader shows information about the state of the battery, battery warnings and errors. Please check information from the display regularly and follow instructions shown on the panel. **Please see the Operator's manual of the loader for more information.**

8. SAFETY FEATURES OF THE BATTERY MANAGEMENT SYSTEM

Battery management system (BMS) monitors battery state and keeps the battery in safe state. The BMS can reduce discharge and charge power to keep the battery in safe state. In some situations, the BMS will disconnect battery from Loader to protect the battery.

In case of over current (charge or discharge), the BMS will first ask the Loader or charger to reduce current. If current is not reduced enough, BMS will disconnect battery from Loader. To reconnect battery, turn the Loader ignition key to OFF for at least five seconds and then turn ignition back to ON.

In case of too high temperature, the BMS will first ask the Loader or charger to reduce power. If temperature keeps rising, BMS will disconnect battery from Loader. To reconnect battery, turn the ignition key to OFF and give time for the battery to cool down and then turn ignition back to ON.

If battery temperature is too low, BMS does not allow connect battery to the Loader. In this case, keep loader ignition key ON and battery will warm itself. If possible, connect Loader to a charger, this will speed up battery warming and reserve battery energy. After battery is warmed reaching operational limits, BMS will connect battery to the Loader.

NOTICE

Note that battery power is still reduced until battery has warmed to above 10°C

In case of deep discharge (state of charge 0 %), BMS will disconnect battery from the Loader. To reconnect battery, it needs to be charged. To prevent deep discharge, never use machine long when battery charge level is close to 0%.

NOTICE

BMS protects battery from over-charging. Over-charging is a rare situation and means that used charger is broken. In normal situation charger will follow battery charge commands and over-charging cannot happen. **Always use the onboard charger of the loader or Avant rapid charger to charge battery.**

9. ROUTINE INSPECTIONS BEFORE USE OF THE E-LOADER

Following inspection guideline should be performed before normal use of the e-Loader.

Table 2 – Inspection guideline

Inspection point	Requirement	Action
Connection cables and connectors	Must be clean, there must not be signs of wear or damage	Contact service to replace any damaged components
Battery mounting space condition	Must be free from water, dirt, dust, or other external objects	Remove with a vacuum cleaner or another suitable device
Battery housing and cover	Must not be damaged	Contact service to replace any components
Rear frame of the loader	Must not be damaged	Contact service to replace any components
Battery is fixed into the mounting space	Fixing must be firm, no movement of the battery is allowed	Contact service if battery can move
If subjected to strong oxidizing agents or acids	Not allowed	Remove and clean with wet wipers or similar
Fire precautions	Fire extinguisher of type AVD/10kg recommended, but at least ABC/10 kg must be available and accessible near the charging area	Provide such extinguisher before use of the Loader. Train how to use the extinguisher

10. MAINTENANCE AND SERVICE

NOTICE

Do not perform other actions except those listed in chapter 9. Routine inspections. Only Avant service is authorized to perform other maintenance actions. Contact your local Avant service provider for further service and maintenance.

11. WARRANTY TERMS

Avant Power Oy grants a warranty of three years (36 months) or 900 charging cycles which ever comes first, from the date of purchase of the battery or the loader.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is performed at the factory or local service dealer.
- The factory or local service dealer replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal wearing of the components
- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Power, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

The warranty labels:

- The warranty labels are located on the steel cover of the battery, see picture 3 and picture 5. The label ensures the information that the cover is removed only by Avant service personnel.
- If one or both of the labels are broken, the warranty is void.

12. EC DECLARATION OF CONFORMITY

The Battery System is a part of the CE marking of the loader according to the applicable regulations.

12.1 Norms & regulations

- MD 2006/42/EC machine directive (EU)
- EMC 2014/30/EC electromagnetic compatability (EU)
- BD 2006/66/EC Battery Directive (EU)
- REACH No 1907/2006 (EU)

Standard	Status
IEC 62619	Fulfilled
UN 38.3 7th Ed.	Fulfilled
IEC 62281	Fulfilled
IEC 62485-6	Fulfilled
EN ISO 12100	Fulfilled
EN ISO 13849	Fulfilled

13. TRANSPORT INFORMATION

The battery is always transported as part of the e-Loader. If the battery is transported separately following rules must be followed:

- Safety documentation must be included.
- After transportation, check that the delivery is complete and not damaged.
- Replacement or separate battery components should only be transported in coordination with dedicated service following the mandatory rules about packaging, labelling and documents which apply for each transport method.

Transport body	UN Number	UN shipping name	Hazard class	Packing group
ADR/RID	UN 3480	Lithium-ion batteries	9	II
IMDG	UN 3480	Lithium-ion batteries	9	II
IATA/ICAO	UN 3480	Lithium-ion batteries	9	II
IMDG	UN 3171	Battery-powered vehicle	9	Battery attached in Loader

Recommendations for safe transportation

If possible, reduce the Li-ion battery charging level to <50% before road transports. The risk in case of an accident will be reduced in relation to the stored energy level. The ambient temperature range during transportation should be within -20°C to +40°C. If the battery temperature is below 0°C after transportation, loader needs to be stored in indoor temperature to acclimatize the battery before normal use. If the vehicle is used with a battery temperature between 0 to 10°C, the performance and functions are limited to protect the cold battery. Below battery temperature 0°C the machine can't be started.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

Not applicable.